

Central research question. Can an agent learn unfamiliar environments efficiently by treating actions as scientific interventions, maintaining competing explanations, and explicitly modeling the mechanisms that produce observed changes?

Motivation

- Pattern recognition may predict outcomes without discovering mechanisms.
- Purely symbolic systems can be brittle when perception is uncertain.
- Passive learning cannot efficiently resolve ambiguity in novel interactive environments.
- Agents must learn what actions mean, infer goals, distinguish correlation from mechanism, and plan with incomplete knowledge.

NEURAL REPS · SYMBOLIC REASONING · ACTIVE EXPERIMENT

Foundational transition — Before → Action → After

$$\mathbf{T}_t = (\mathbf{s}_t, \mathbf{a}_t, \mathbf{s}_{t+1}) \quad \Delta_t = \text{Diff}(\mathbf{s}_t, \mathbf{s}_{t+1})$$

$\mathbf{s}_t$ structured state before · $\mathbf{a}_t$ action/intervention · $\mathbf{s}_{t+1}$ resulting state · Δ_t observed effect.

Transitions provide evidence about action semantics, environmental rules, causal mechanisms, invariants, goals, and terminal conditions.

Fig. 1 — Transition evidence unit for mechanism learning.

Object-centric reasoning

State as entities and relations — not only pixels or a global novelty vector.

Object: identity, type, color, shape, position, attributes, history
Relations: adjacent-to · inside · aligned-with · blocking · approaching · connected-to

$$\text{Effect}(\mathbf{a}, \mathbf{o}_i, \mathbf{x}_{ij}, \mathbf{c}_t) \rightarrow \Delta \mathbf{o}_i$$

Detected objects and relations must influence hypothesis generation, transition prediction, experiment selection, action scoring, and planning.

Fig. 2 — Binding interventions to object–relation structure.

Memory

Episodic (runs) · transition (before–action–after) · semantic (concepts/rules) · procedural (successful sequences) · failure (contradicted rules) · experiment (hypotheses, configs, results, conclusions).

Architecture

ASRA-NFM architecture flowchart

Fig. 3 — Observation → perception → state → intervention → transition analysis → hypotheses → hybrid world model → information gain → planning → next action, with feedback into memory and belief revision. Styling: neural (cyan), symbolic (amber), evidence (green), planning (violet).

Tabby — Questions

Identifies uncertainty about actions, mechanisms, goals, objects, relations; proposes discriminating questions and interventions. Functional reasoning module — not necessarily a separate language model.

Grubby — Evidence

Answers from recorded observations and transitions; support, contradictions, invariants, and counterexamples.

Sam — Decisions

Chooses whether to explore, test a hypothesis, or execute a goal-directed plan.

Hypothesis management

$$\mathbf{H}_t = \{h_1, h_2, \dots, h_k\}$$

$$\text{Score}(\mathbf{h}) = \text{Accuracy}(\mathbf{h}) + \alpha \text{Coverage}(\mathbf{h}) - \beta \text{Complexity}(\mathbf{h}) - \gamma \text{Contradictions}(\mathbf{h})$$

H1 ACTION_2 moves every red object. **H2** moves the object adjacent to the agent. **H3** moves the selected object when the path is clear.

Prefer hypotheses that predict accurately, explain more evidence, remain simple, and survive attempted falsification. Interventions distinguish these possibilities.

Fig. 4 — Competing falsifiable mechanisms.

Active experimentation

$$\mathbf{a}_t^* = \text{argmax}_{\mathbf{a}} [\text{IG}(\mathbf{a}; \mathbf{H}_t) + \lambda \text{P}(\text{progress} \mid \mathbf{a}) - \mu \text{Risk}(\mathbf{a}) - \nu \text{Cost}(\mathbf{a})]$$

An action is valuable if it advances the inferred objective or reveals which world-model hypothesis is correct.

Applications & methodology

Testbed: interactive ARC-AGI-style environments. **Broaders:** novel envs, autonomous ML experimentation, scientific discovery, robotics, biological decision systems, physical AI. Adapters share one adaptive scientific-reasoning core.

Baselines: random · heuristic · neural-only · symbolic-only · hybrid · hybrid + active IG.

Ablations: –object relations · –hypothesis memory · –neural WM · –active experiment · –failure memory · –complexity penalty.

Metrics: success, score, action efficiency, prediction accuracy, mechanism recovery, goal inference, IG/action, calibration, compute, generalization.

Results Preliminary result; controlled evaluation in progress

Current benchmark: [BENCHMARK] · Score: [SCORE] · Previous: [PREVIOUS_SCORE] · Δ: [DELTA]
Environments N: [N] · Date: [DATE] · Version: [VERSION] · Hardware: [HARDWARE]

Placeholder evaluation charts

Fig. 5 — Editable SVG charts. Dashed bars mark missing measurements. Do not interpret as empirical claims.

Main research claim (hypothesis)

An object-centric neuro-symbolic agent with explicit hypothesis management and active intervention can learn novel mechanisms more accurately and with fewer actions than passive, neural-only, and symbolic-only baselines.

Contributions

- Transition-centered learning from before–action–after evidence
- Explicit separation of action semantics, mechanisms, and goals
- Object-centric causal representation
- Competing, falsifiable symbolic hypotheses
- Hybrid neural + symbolic world modeling
- Information-seeking intervention selection
- Interpretable evidence and decision traces
- Reusable architecture for interactive reasoning and scientific experimentation